Proof. For an RV-valued f, put $h = \mathrm{vrv} \circ f$; for a Γ-valued f, put $h = f$. We retain the induction on k, with the case $k = 0$ immediate. For $k > 1$ and $t \in \mathrm{pr}_1(U)$, the inductive hypothesis makes $h(U_t)$ doubly bounded. Its finite bound

$$\gamma_t = \sup\{|h(t, x)| \mid x \in U_t\}$$

exists by definable completeness of Γ, and the least-upper-bound formula makes $t \longmapsto \gamma_t$ definable. The projection $\mathrm{pr}_1(U)$ is doubly bounded, so the unary Γ-valued case bounds all γ_t and hence bounds $h(U)$. Thus both claims reduce to the case $k = 1$ and a definable function $g : U \longrightarrow \Gamma$. Then, by C-minimality, we may assume that g vrv-contracts to a function $g_\downarrow : \mathrm{vrv}(U) \longrightarrow \Gamma$. Since definable functions in Γ are piecewise $\mathbb{Q}$-linear, the range of $g_\downarrow$ must be doubly bounded. □

Resolved referee R037 [Gap] – Uniformity in the bounded RV-image lemma.
Original source line 1017. The induction on k must control the bounds uniformly in the remaining coordinates; boundedness on each fiber alone does not give a bound on their union. A precise route is to consider the relation of possible values $\mathrm{vrv}(f(u))$ over $\mathrm{vrv}(u) \in \Gamma^k$. Residue-field/value-group orthogonality makes each such fiber finite, uniformly; stable embeddedness and elimination in divisible ordered abelian groups then give finitely many piecewise rational-affine branches. These have bounded images on a bounded box. Include this justification, or an equivalent explicit uniform induction.

Author resolution. The author retains the original induction. Its fiber bounds are chosen definably as finite suprema of the absolute output values; the unary Γ-valued case then bounds this definable bound function uniformly. The proof now spells out that step, without replacing the induction by a relation of all input and output values.

Definition 2.42. Let $A \subseteq \mathrm{VF}^n$ and $f : A \longrightarrow \mathcal{P}(\mathrm{VF}^m)$ be a definable function. For $a \in \mathrm{VF}^n$ and $L \subseteq \mathrm{VF}^m$, we say that L is a *limit set of f at a*, written as $\lim_{A \to a} f \subseteq L$, if for every $\epsilon \in \Gamma$ there is a $\delta \in \Gamma$ such that if $c \in \mathfrak{o}(a, \delta) \cap (A \setminus \{a\})$ then

$$f(c) \subseteq \bigcup_{b \in L} \mathfrak{o}(b, \epsilon).$$

This is a local definition and imposes no global boundedness hypothesis. A finite limit set forces the union of the values $f(c)$ to be bounded on some punctured neighborhood of a. Whenever boundedness of the range is assumed, it means boundedness of $\bigcup_{c \in A} f(c)$.

A limit set L of f at a is *minimal* if no proper subset of L is a limit set of f at a.

Resolved referee R038 [Clarification; revised] – Simplify the limit-set definition.
Original source line 1046. Say explicitly that $\bigcup_{a \in A} f(a)$ is bounded. The auxiliary subset L' does no work: the displayed condition is equivalent to $f(c) \subseteq \bigcup_{b \in L} \mathfrak{o}(b, \epsilon)$. The following definition of continuity is applied to finitary functions without a global boundedness hypothesis, so the present definition needs a compatible extension. One can use the same formula for arbitrary correspondences whenever a finite affine limit exists, or formulate limits in $(\mathbb{P}^1(\mathrm{VF}))^m$ from the outset; local boundedness near the point suffices for affine local uses.

Author resolution. The author approves the local limit formula without a global range bound, using all of L in place of the redundant subfamily L'. A finite limit set supplies local boundedness on a punctured neighborhood. Any separately assumed range bound refers to the union of all values. Minimality and the puncture are unchanged.

Observe that if $\lim_{A \to a} f \subseteq L$ and $b \in L$ is not isolated in L then actually $\lim_{A \to a} f \subseteq L \setminus b$. So in a minimal limit set every element is isolated. Moreover, if a minimal limit set L exists then its topological closure is unique:

Lemma 2.43 ([17, Lemma 9.2])**.** *Let $L_1, L_2 \subseteq \mathrm{VF}^m$ be two minimal limit sets of f at a. Then the closures of L_1, L_2 coincide.*

This lemma justifies writing $\lim_{A\to a} f = L$ when L is a closed (hence the unique) minimal limit set of f at a, in particular, when L is finite and minimal.

Let $A \subseteq \mathrm{VF}^n$ be a definable set with $\dim_{\mathrm{VF}}(A) = n$ and $f : A \longrightarrow \mathcal{P}(\mathrm{VF}^m)$ a definable finitary function. We say that f is *continuous* if $\lim_{A\to a} f = f(a)$ for all $a \in A$.

Lemma 2.44. *Suppose $n > 0$. Then f is continuous almost everywhere.*

Proof. This can be reduced to the familiar case that f is indeed a function into VF^m as follows. By Lemma 2.12 and compactness, there are a definable set U in RV and a definable function $g : A \times \mathrm{VF}^m \longrightarrow A \times U$ such that, for all $a \in A$, $\mathrm{pr}_A(g(a, b)) = a$ and $g \upharpoonright (a \times f(a))$ is injective. For each $t \in U$, let

$$A_t = \{a \in A \mid (a, t) \in g(a \times f(a))\}$$

and $f_t \subseteq f$ be the obvious t-definable function $A_t \longrightarrow \mathrm{VF}^m$. The discontinuity locus E of f is definable: the limit condition is first-order, and minimality of the finite set $f(a)$ can be tested by removing one of its elements at a time. Suppose for contradiction that $\dim_{\mathrm{VF}}(E) = n$. By Lemma 2.17, E contains a nonempty open polydisc. Apply generic local constancy [17, Lemma 8.9] to the finite label-set map

$$a \longmapsto g(a \times f(a))_{\mathrm{RV}}.$$

After shrinking to a nonempty open polydisc $P \subseteq E$, this set is a fixed finite set T. If T is empty, then f is the constant empty correspondence on P and is continuous there, a contradiction.

For each $t \in T$, the inverse branch f_t is defined on all of P. Extend it by zero outside A_t and apply Lemma 2.36. The finitely many branch-exception sets have union of VF-dimension less than n, so choose $a \in P$ outside their union. All branches are continuous at a. Their finite union has limit set $f(a)$; this limit set is minimal because the distinct values at a can be separated by sufficiently small discs and, since $n > 0$, the punctured polydisc still accumulates at a. Thus f is continuous at a, contradicting $P \subseteq E$. Hence $\dim_{\mathrm{VF}}(E) < n$. □

Resolved referee R039 [Gap; revised] – One continuous branch is not enough.
Original source line 1068. The argument finds one t for which A_t has full dimension and f_t is generically continuous. That alone does not account for all branches of f. Apply generic local constancy to the finite RV-label set $a \mapsto g(a \times f(a))_{\mathrm{RV}}$, then simultaneously apply generic continuity to its finitely many inverse branches on each neighborhood. To justify the global dimension conclusion, suppose the discontinuity locus had full dimension and choose an open polydisc in it where this label set is constant; the finite collection of branch-exception sets cannot fill that polydisc. The empty label set simply gives the constant empty correspondence and should be handled separately. This supplies the missing coverage argument.

Author resolution. The author approves the expanded proof and the restriction $n > 0$. The original ending uses the dimension principle that an RV-indexed definable union of sets of VF-dimension less than n still has dimension less than n. The expanded argument obtains the same coverage by making the finite label set locally constant and treating all branches together, including the empty-label case. The positive-arity hypothesis is needed for the unchanged punctured limit convention.

Lemma 2.45 ([17, Lemma 9.5])**.** *Suppose that $f : \mathcal{M}\smallsetminus 0 \longrightarrow \mathcal{P}(A)$ is a definable finitary function, where A is a closed and bounded set in* VF*. Then there is a definable finite set $L \subseteq A$ such that $\lim_{\mathcal{M}\smallsetminus 0\to 0} f = L$.*

We remark that if a "point at infinity" is adjoined in each coordinate (one-point compactification of VF) then there is no need to require that A be bounded.

revision of the next few lemmas